Aditya Nivrutti Jadhav

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Java Software Engineer | xOTA & UDS (ISO 14229) | Embedded Java | AWS Solutions Architect | AI/LLM Integration

Professional Summary

Embedded Java Software Engineer with 3.8+ years building xOTA and UDS (ISO 14229) diagnostic solutions for connected trucks and buses. Expertise in ECU flashing/reprogramming, dual-bank updates, Global xOTA scripting for parameterization, troubleshooting, and vehicle health, test-bench and end-to-end release-pipeline automation using GitHub Actions and Docker. Familiar with SDV architecture and vehicle telematics. AWS Certified Solutions Architect - Associate; M.Tech, NIT Calicut.

Technical Skills

OTA & Flashing: FOTA/POTA/DOTA/TOTA, ECU Flashing/Reprogramming (UDS flash sequence), Dual-Bank (A/B) ECU Updates, Daimler Diagnostic specification, Vehicle Health, SDV Architecture, Vehicle Telematics, Embedded Java, MQTT, Protobuf, ISO 24089, UNECE R156, ISO/SAE 21434
Diagnostics: **UDS (ISO 14229)**, SOVD (ISO 17978), OTX (ISO 13209), ODX (ISO 22901), Can, DoIP, SOME/IP, Automotive Ethernet, AUTOSAR, DTCs/NRCs, Daimler Diagnostic Sequences, CANoe, DiagnosticLink, ISO 26262
Languages & Backend: Java, Python, C/C++, SQL, Spring Boot, JUnit, Maven/Gradle, GraalVM Native Image, REST APIs, Microservices, Design Patterns
Cloud & Delivery: AWS (EKS, VPC, S3, ECS), Docker, Kubernetes, GitHub Actions, CI/CD
AI/GenAI: RAG, LLM Integration (Claude, OpenAI), Embeddings, Self-Hosted LLMs (Qwen, Llama), Agent Orchestration

Professional Experience

Daimler Truck Innovation Center India — Bengaluru, India
Software Engineer – xOTA & Vehicle Diagnostics — *Mar 2024 – Present*

- Develop and own selected features of an embedded Java-based xOTA solution for Daimler trucks and buses, including Global xOTA scripts that orchestrate UDS (ISO 14229)-based ECU flashing/reprogramming, parameterization, and troubleshooting across 1M+ connected devices and 5+ products.
- Develop and maintain Global xOTA scripts for passive-inventory jobs that read ECU data items and VCD IDs for vehicle-health products. Work with UDS (ISO 14229) diagnostic/ECU-flashing sequences and a proprietary diagnostics DSL comparable to OTX (ISO 13209); built a Python static-analysis tool that validates scripts before execution, reducing code-review time by 30%.
- Designed a backward-compatible integration layer for legacy automotive platforms, preserving existing interfaces while enabling system evolution and compatibility across platform versions.
- Contributed to migrating a Java application to GraalVM Native Image, reducing startup time from 2–3 minutes to approximately 40 seconds compared with traditional JVM/JIT startup.
- Implemented GitHub Actions-based release automation for packaging and deployment, reducing deployment time by 40% and manual release errors by 90% across 4 teams.
- Automated sequence-driven diagnostic test-bench operations, reducing manual testing time by 50%, and built a RAG-based code-review and recurring-defect prediction tool using Jira and Bitbucket data, achieving 80% prediction accuracy for recurring defects and supporting early risk detection for 50+ engineers.
- Deployed AI engineering tooling on Amazon EKS within an AWS VPC using self-hosted Qwen and Llama models and embeddings-based retrieval to keep proprietary source code and defect data within a private environment.

Tata Consultancy Services (TCS) — Pune, India
Systems Engineer – Enterprise Platform Integrations — *Jun 2021 – Jul 2022*

- Managed a ServiceNow and SharePoint migration for a 500+ user enterprise client, coordinating configuration, data migration, and cutover across 3 cross-functional teams with zero downtime.
- Developed 5+ ServiceNow inbound actions and REST API integrations, validated end to end using Postman, and resolved 20+ integration defects to support stable production releases.

Education

National Institute of Technology Calicut — Kerala, India
M.Tech in Computer Science and Engineering — *Aug 2022 – Mar 2024 | CGPA: 8.62/10*

Dr. J. J. Magdum College of Engineering, Jaysingpur — Maharashtra, India
B.E. in Computer Science and Engineering — *Aug 2017 – Jun 2021 | CGPA: 8.40/10*

Research & Publications

M.Tech Thesis, NIT Calicut – Minimum Link Watchman Route Problem — 2024

- Analyzed the computational complexity of route-planning variants on geometric arrangements, producing formal proofs and a published paper.

Certifications & Achievements

AWS Certified Solutions Architect – Associate (SAA-C03)

Microsoft Azure Fundamentals (AZ-900)

IBM Data Science Methodology and **Machine Learning with Python (IBM)**

GATE CSE: **96.37 percentile**; solved **500+ DSA problems** on LeetCode and GeeksforGeeks.